

Unit-1: .NET Framework Basics

1. .NET Comparison & Overview

- Frameworks: **.NET Framework, .NET Core, .NET 5+** 
- **C#, VB.NET, F#** supported
- Cross-platform (.NET Core/.NET 5+) 
- Common Features: CLR, BCL, CTS, CLS, MSIL

2. CLR Architecture

- **Common Language Runtime** → Executes .NET apps 
- Components: **JIT, Garbage Collector, Class Loader, Type Checker** 
- Handles **memory, security, exception, threading**

3. JIT (Just-In-Time Compiler)

- Converts **MSIL → Machine Code** at runtime 
- Types:
 - **Pre-JIT** → Compiles entire code at once
 - **Econo-JIT** → Compiles only needed methods
 - **Normal JIT** → Compiles methods when called

4. MSIL, CTS, CLS, BCL

- **MSIL**: Intermediate language 
- **CTS**: Standard data types across languages 
- **CLS**: Rules for language interoperability 
- **BCL**: Pre-built classes for common tasks 

5. Advantages of .NET

- Multi-language support, memory management 
- Cross-platform 
- Secure , fast , reusable & maintainable 

Unit-2: OOPS & Toolbox

1. Advantages of OOPS

- Reusability , Maintainability 
- Scalability , Security 

- Real-world modeling 🌐, Polymorphism & Abstraction ✨
2. **Exception & String Handling**
 - **Exception:** try-catch-finally ⚠️
 - **String Methods:** Length, Substring, Replace, ToUpper, Trim ✍️
 3. **Toolbox Controls**
 - Buttons 🕒, Labels 🏷️, TextBox ✍️
 - CheckBox ✅, RadioButton ♂, ListBox 📋, ComboBox ⬇️
 - PictureBox 🖼️, ProgressBar ⌚, Timer ⌚
 4. **Property & Event Explorer, Debugging, Class & Command View**
 - Property → Modify control properties 🎨
 - Event → Handle events ⚡
 - Debug → Breakpoints, Watch, Step ⚙️
 - Class View → Navigate classes 📖
 - Command → Execute commands 💻
 5. **Menu Bar Control**
 - MenuStrip → File, Edit, View 📋
 - ToolStripMenuItem → Menu items
 - Event Handling → Click actions ⚡
 - Shortcut Keys → Ctrl+S etc. ⌨️
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Unit-3: Forms & Controls

1. **MDI Form**
 - Parent form with multiple child forms 📁
 - Useful for document-based apps ✨
2. **Common Dialog Box**
 - OpenFileDialog 📁, SaveFileDialog 💾
 - ColorDialog 🎨, FontDialog ✍️, PrintDialog 🖨️
3. **Common Controls (10)**
 - Button 🕒, Label 🏷️, TextBox ✍️
 - CheckBox ✅, RadioButton ♂, ListBox 📋

- ComboBox , PictureBox , ProgressBar , Timer 

4. Month Calendar & Date-Time Picker

- Month Calendar  → select dates
- Date-Time Picker  → select date/time

5. CheckBox & ComboBox

- CheckBox → multiple options 
- ComboBox → single dropdown 

6. Message Box & Input Box

- MessageBox → Alerts, info 
- InputBox → Get user input 

Unit-4: ADO.NET & Data Handling

1. Architecture of ADO.NET

- Data Provider , DataSet , DataAdapter 
- Supports **connected & disconnected mode** ⚡

2. Crystal Report & Integration

- Create formatted reports 
- Steps: Install → Add Report → Design → Connect → ReportViewer → Load in code

3. Data Adapter, Data Provider, DataSet

- DataProvider → Connection, Command, DataReader
- DataAdapter → Fill/Update DataSet 
- DataSet → In-memory tables, relations 

4. Execute Methods

- ExecuteNonQuery → INSERT/UPDATE/DELETE 
- ExecuteScalar → Single value 
- ExecuteReader → SELECT, forward-only 

5. Connection

- Connect app to DB 
- SqlConnection, OleDbConnection, OracleConnection
- Open(), Close(), Transaction support 